

Yohei Ohata Curriculum Vitae

Center for Information and Neural Networks
National Institute of Information and Communications Technology
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Education

The University of Tokyo

April 2022 – March 2024

M.S. in Human Factors Engineering

Department of Human and Engineered Environmental Studies

Cumulative GPA: 4.0/4.0

Thesis: Identifying Different Thought Processes for Each Learner Using Eye-Tracking and Hidden Markov Models

1st in Class (1/48), Dean's Award, Best Department Thesis Award

Tokyo University of Science

April 2018 – March 2022

B.E. in Electrical and Electronic Engineering

Department of Applied Electronics

Cumulative GPA: 3.5/4.0

Thesis: A Method of Designing a Hilbert Transformer with Transmission Zeros at Specified Positions

Employment

National Institute of Information and Communications Technology

January 2026 – Present

Research Assistant

Center for Information and Neural Networks (CiNet)

Computational Social Neuroscience Group

Hitachi, Ltd.,

April 2024 – December 2025

Research Engineer

R&D Group

Publications

Articles in Peer-Reviewed Journals

J1 Ohata, Y., Hachisuka, S., Kurita, K., & Warisawa, S. (submitted). Identifying Different Thought Processes for Each Learner Using Eye-Tracking and Hidden Markov Models. .

Research Experience

Research Assistant

January 2026 – Present

National Institute of Information and Communications Technology

Laboratory: Computational Neuroscience & Decision Making Laboratory

Supervisor: Haruno Masahiko

- Trained a toy Transformer model on random walks over a community-structured graph and visualized the internal representations as a Hopfield energy landscape.
- Elucidating shared and distinct computational principles underlying hippocampal memory and the attention mechanism.

Research Assistant

September 2025 – Present

The University of Tokyo

Laboratory: Amano & Nakayama Lab

Supervisor: Yamashita Ayumu, Amano Kaoru

- Performed cross-subject temporal synchronizing on the fMRI dataset (AX-CPT, Cued Task Switching, Sternberg, Stroop) using BrainSync (MATLAB).
- Applied tensor decomposition to the DMCC55B dataset to decompose data into spatial, temporal, and subject-load modes (MATLAB).
- Quantified cognitive control performance by indexing RT and Accuracy via the Cognitive Control Index, extracting contributory components using conditional randomization testing (R).

Research Assistant

April 2022 – March 2024

The University of Tokyo

Laboratory: Inovative Learning Creation Lab

Supervisor: Hachisuka Satori, Warisawa Shin'ichi

- Designed experimental protocols for both adult and elementary school participants, including the development of a web-based UI and logging system using HTML/CSS/JS.
- Managed high-fidelity eye-tracking data collection and processing using Tobii Pro 3 and Tobii Pro Lab.
- Performed advanced statistical analysis using Linear Mixed-Effects Models (LMM) in R.
- Modeled the temporal dynamics of scan paths using Hidden Markov Models (HMM) implemented in Python.
- Conceptualized research goals, authored manuscripts.

Professional Experience

Teaching Experience

Sensing and Signal Processing

April 2023 – June 2023

Teaching Assistant, The University of Tokyo

- Assisted in building signal processing systems using LabVIEW

Honors and Awards

IEEE CAS JJC Best Student Award

2021

Languages

- High Level in English
- Native in Japanese

References

Haruno Masahiko, Ph.D.,

Professor, National Institute of Information and Communications Technology

Email: mharuno@nict.go.jp

Ayumu Yamashita, Ph.D.,

Associate Professor, Kobe University

Email: ayumu722@people.kobe-u.ac.jp

Robert Mok, Ph.D.,

Principal Investigator, National Institute of Information and Communications Technology

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Satori Hachisuka, Ph.D.,

Associate Professor, The University of Tokyo

Email: hachisuka@edu.k.u-tokyo